HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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Formal verification
SEREs
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Timer

Using the timer exercise, write PSL assertions that describe its behavior with SEREs.
You may also write simpler assertions similar to the LTL-style properties from the course, or explore longer sequences that combine several behaviors.
For instance, it should be possible to describe what is happening to the timer from the moment start is active. Define a named SERE for a scenario that starts when `start_i` is asserted and continues through timer completion.

Elevator

Using the elevator exercise, write PSL assertions that describe its behavior with SEREs.
For example, in elevator version 2, define a named SERE for a scenario in which the elevator is at floor 0 with its door closed, the floor 1 call is asserted, the elevator travels to floor 1, and `open_o` is asserted for the specified interval. Use the sequence in an assertion to check the required behavior and in a `cover` directive to check that the scenario is reachable. Keep these goals distinct : reaching the sequence does not prove that the asserted behavior is correct.
You could also define and cover a complete trip : the elevator starts on floor 0, travels to floor 1, then returns to floor 0. Reuse named SEREs where useful in both assertions and cover directives. These are suggested scenarios ; you may explore other useful sequences as well.